DATA 266 Homework 8

Mini Diffusion Model DDPM and DDIM

Siva Surya Chandran | SJSU ID 019130215 | Seed 3215

Main conclusion. The model learned the noise-prediction task, but the assigned 100-step schedule leaves a noticeable amount of signal at the final timestep. DDIM sampling is much faster and deterministic when eta equals zero, while DDPM remains stochastic. The results below use the latest recorded run.

1 Overview

This homework implements a small diffusion model on MNIST. The forward process gradually adds Gaussian noise to handwritten digits. A time-conditioned convolutional neural network learns to predict that noise, and the learned model is then used to reverse the process with DDPM and DDIM sampling.

The assignment configuration uses 100 diffusion steps, a linear beta schedule from 0.0001 to 0.02, a batch size of 128, 50 training epochs, and Adam with a learning rate of 0.0002. The network has five convolutional layers and 416,641 trainable parameters.

2 Forward diffusion

The forward process adds Gaussian noise to an image. The closed-form equation lets me create a noisy image at any timestep without simulating every earlier step:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1 - \bar{\alpha}_t)}\, \varepsilon, \quad \varepsilon \sim N(0, I)$$

The visualization shows that the digit is still visible at early timesteps and becomes heavily corrupted later. At t = 100, a faint trace can remain because the schedule is relatively short. In this run, $\bar{\alpha}_{100}$ is about 0.364, so $\sqrt{\bar{\alpha}_{100}}$ is about 0.60. The final state is therefore not exactly a standard Gaussian, and starting the reverse process from N(0, I) is an approximation.

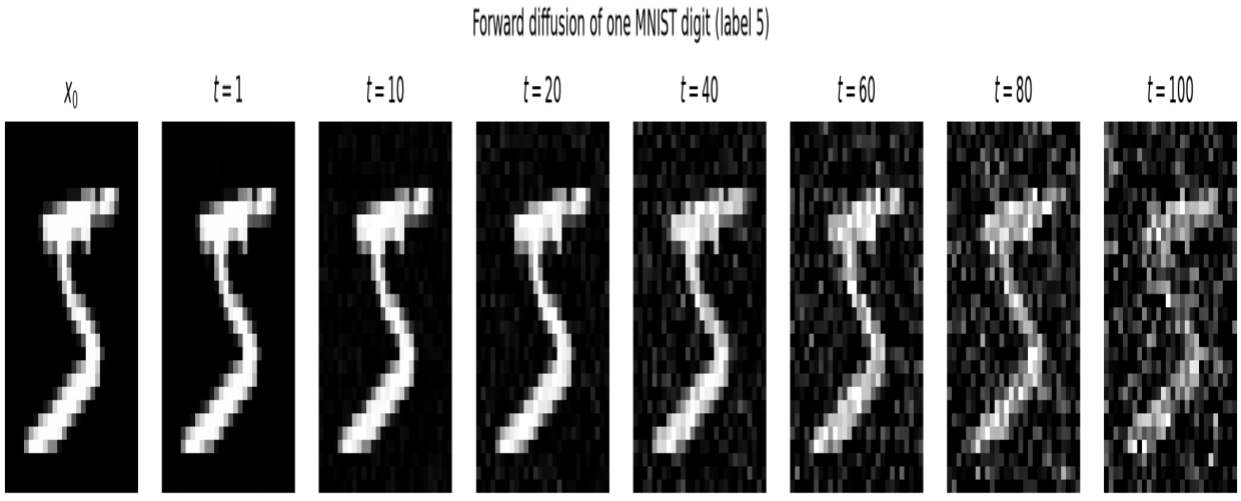


Figure 1. Forward diffusion of one MNIST digit.

3 Model and training

The network receives a noisy image and its timestep. A sinusoidal timestep embedding is passed through a small multilayer perceptron and added inside the convolutional blocks. SiLU nonlinearities and a skip connection help the network preserve useful image structure while processing different noise levels. The output has the same 1 × 28 × 28 shape as the input.

Training uses the simple noise-prediction loss, which measures the mean squared error between the true noise and the network's prediction. The latest run ended with a final epoch loss of 0.07246 after about 42.6 minutes on Apple MPS.

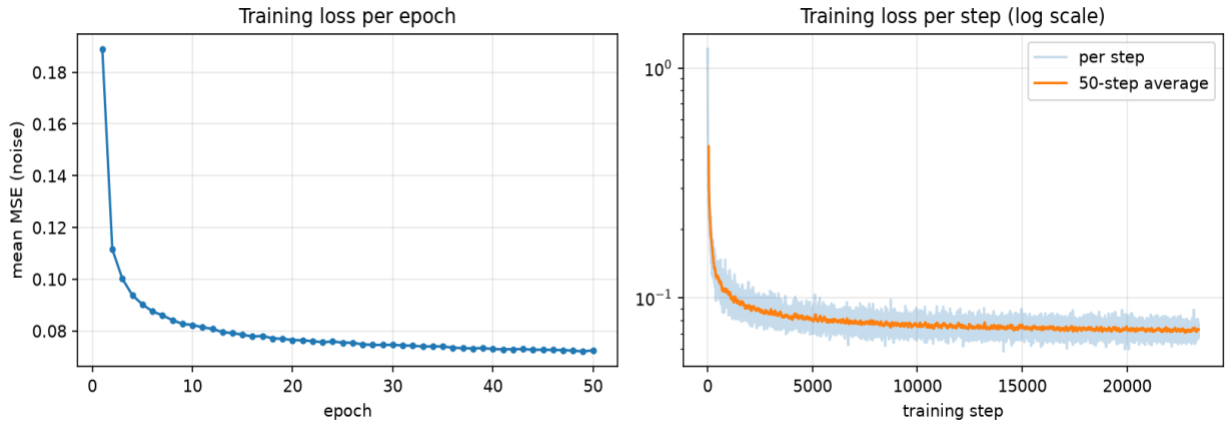


Figure 2. Training loss during the latest run.

4 DDPM sampling

DDPM sampling starts with Gaussian noise and steps backward through all 100 timesteps. The network prediction is used to compute the reverse-process mean. I used the standard DDPM posterior variance:

$$\tilde{\beta}_t = \beta_t \, (1 - \bar{\alpha}_{t-1}) \, / \, (1 - \bar{\alpha}_t)$$

Fresh random noise is added at each reverse step except the final step. This makes DDPM stochastic: changing the reverse-noise seed changes the generated output even when the starting state is the same.

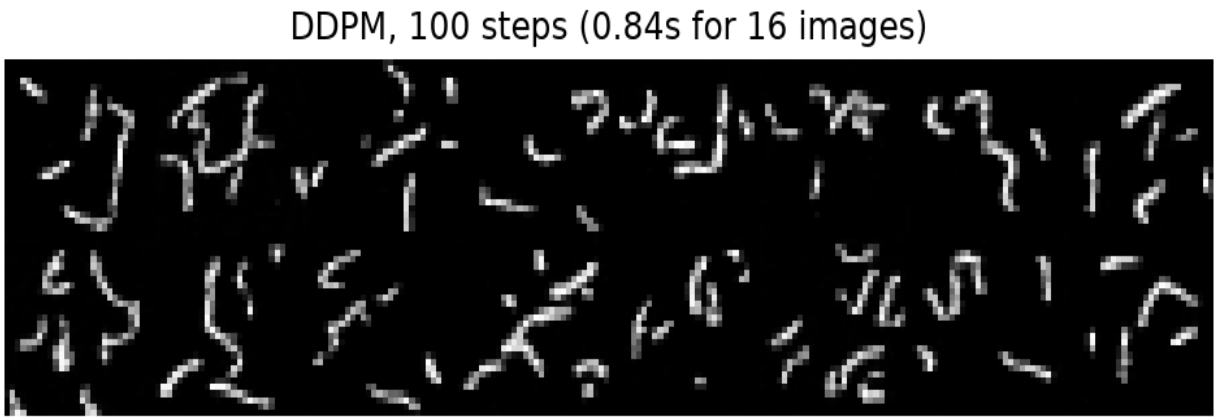


Figure 3. DDPM samples from the latest recorded run.

5 DDIM sampling

DDIM uses the same trained network but can skip timesteps. With eta equal to zero, the random part of the update is removed, so the same starting noise produces the same result. I compared DDIM with 100, 50, and 20 steps.

Fewer steps reduce the number of network evaluations and make sampling faster. The trade-off is that larger jumps can change the appearance of the strokes and accumulate more approximation error. With the full 100-step schedule, eta equal to one has the same posterior variance as standard DDPM. It should not be called exactly identical here because the DDIM path clips its estimate of x_0 before the update.

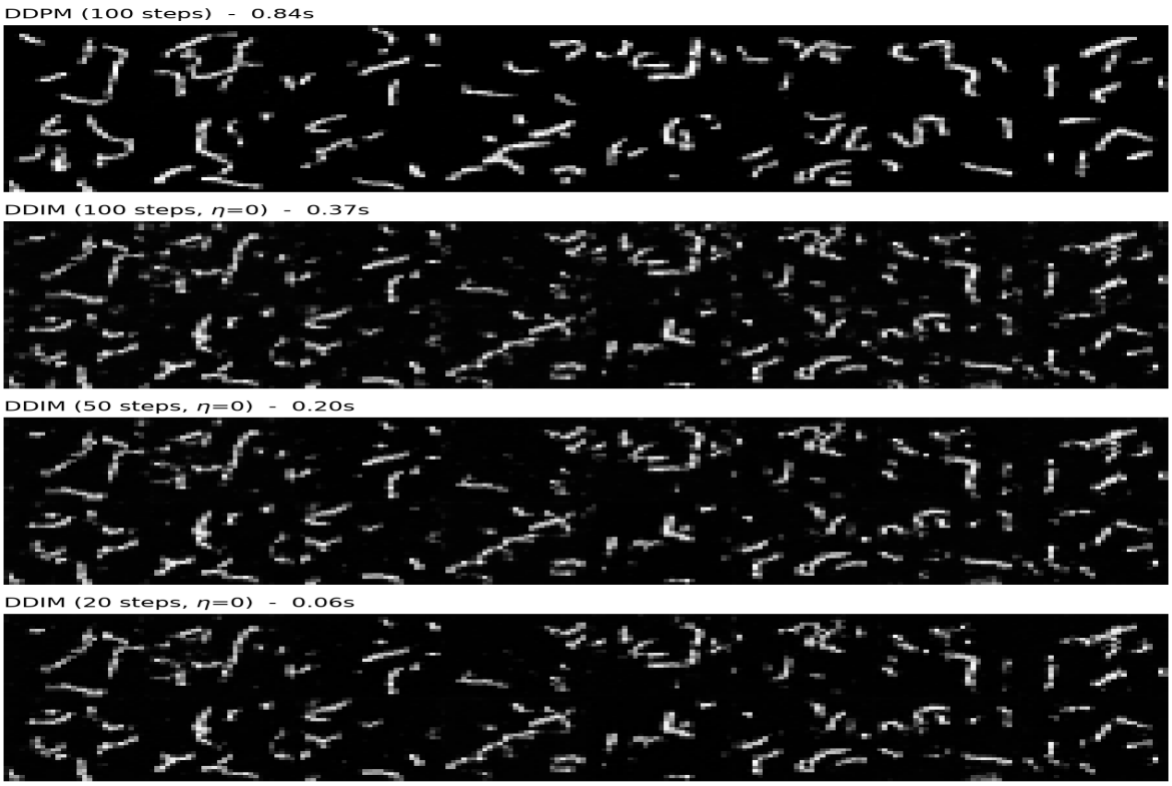


Figure 4. DDPM and DDIM samples using the same trained network.

6 Latest results

The table uses the latest values saved in outputs/results.json. Confidence is a rough proxy from a small MNIST classifier, not a complete measure of sample quality.

Sampler	Evaluations	Time for 256 (s)	Confidence	Classes
DDPM 100	100	7.59	0.6809	10
DDIM 100 eta 0	100	6.59	0.6309	9
DDIM 50 eta 0	50	3.82	0.6467	9
DDIM 20 eta 0	20	1.39	0.6470	9
Real MNIST reference	—	—	0.9897	—

DDIM-50 took about half the time of DDPM-100, and DDIM-20 took about one fifth of the time. Repeating DDIM-50 from the same starting noise gave a maximum absolute difference of 0.0. DDPM remained stochastic; changing its reverse-noise seed gave a mean absolute difference of about 0.0656.

7 Discussion and limitations

The main difference between the samplers is the reverse update. DDPM uses every timestep and injects random noise, while DDIM can skip steps and can be deterministic. This makes DDIM useful when faster sampling is more important than reproducing the full stochastic reverse chain.

The schedule is an important limitation of this experiment. With only 100 steps and a final beta of 0.02, the forward process does not completely remove the original signal. The final forward distribution is therefore not exactly N(0, I), even though the reverse process starts there. This mismatch can make generation harder and should be considered when interpreting the samples.

The classifier confidence should also be interpreted carefully. It comes from a small classifier trained for only three epochs, does not fully measure diversity, and is based on one random seed. A longer schedule, a larger model, or a starting distribution better matched to the final forward state could improve the generated images.

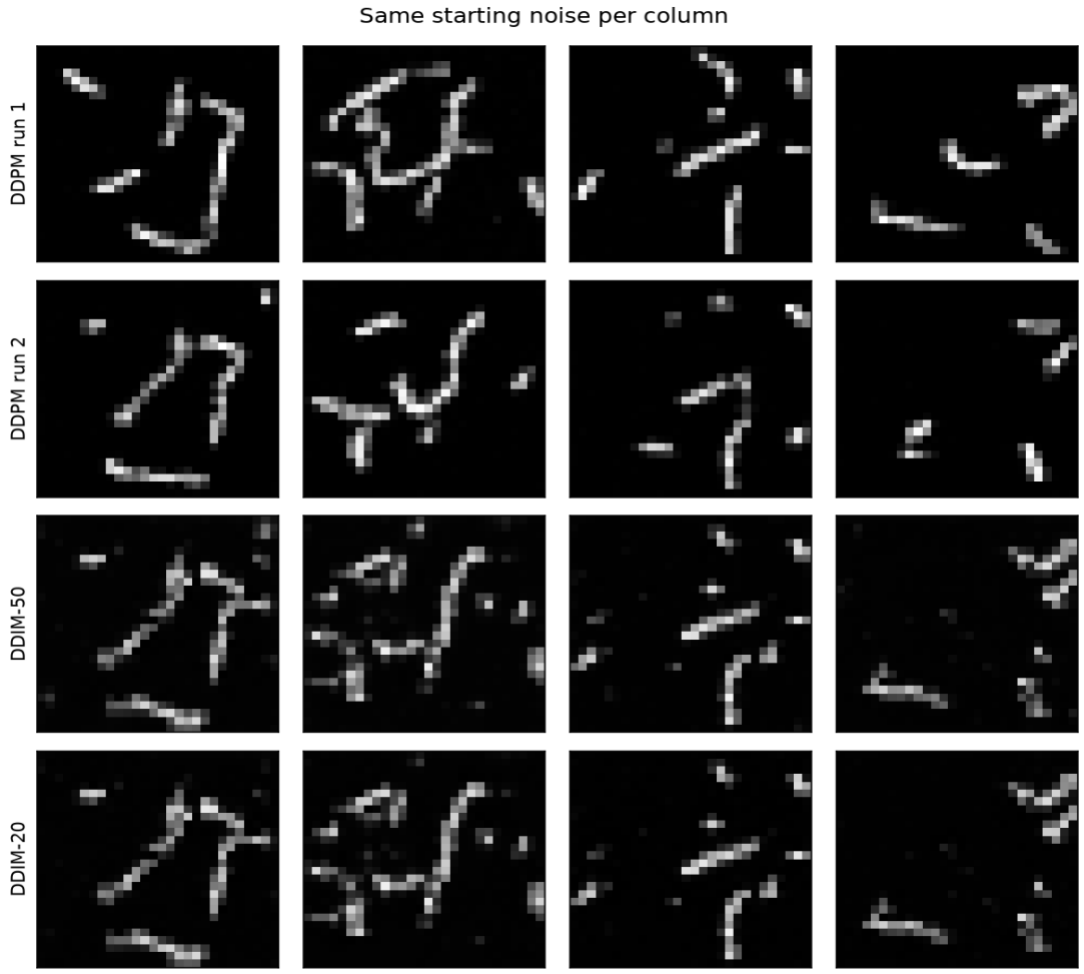


Figure 5. DDPM is stochastic, while DDIM with eta equal to zero is deterministic from the same starting noise.